48V in / 12V out / 150W
DC-DC Converter Module
Model Number
V48C12C150A

Features

- DC input range: 36 - 75V
- Input surge withstand: 100V for 100ms
- DC output: 12V ±1%
- Programmable output: 10 to 110%
- Regulation: ±0.1% no load to full load
- Efficiency: 86%
- Maximum operating temperature: 100°C at 150W
- Power density: 90W/cubic inch
- Height above board: 0.43 in. (10,9 mm)
- Parallelable, with N+M fault tolerance
- PAC (phased array control) compatible
- Low noise ZCS/ZVS architecture



Shown actual size:
2.28 x 1.45 x 0.5 in
57.9 x 36.8 x 12.7 mm

Typical Applications: telecommunications and distributed power systems.

Product Overview

This DC-DC converter module uses second generation power processing, control and packaging technologies to provide the performance, flexibility and cost effectiveness expected of a mature Power Component. For example, a plated-cavity core couples widely separated primary and secondary windings, resulting in low in-to-out parasitic capacitance and noise.
High frequency ZCS/ZVS switching, advanced power semiconductor packaging and thermal management provide high power density with low temperature gradients. Extensive use of silicon integration results in 1/3 the part count of a first generation converter.

This converter is manufactured in a state-of-the-art automated manufacturing facility on a short-cycle-time robotic line having a capacity of one module every ten seconds. In-line process controls are designed to achieve low defect rates and consistent quality. A comprehensive CIM system controls the dispatching, assembly and testing of each module, enabling small lots to be manufactured as efficiently as hundreds-of-thousands of units.

Absolute Maximum Ratings

Parameter	Rating	Unit	Notes
+In to –In voltage	–0.5 to 75	Vdc	
+In to –In voltage	100	Vdc	<100ms
PC to –In voltage	–0.5 to 7.0	Vdc	
PR to –In voltage	–0.5 to 7.0	Vdc	
+Out to –Out voltage	–0.5 to 20	Vdc	
SC to –Out voltage	–0.5 to +1.5	Vdc	
Isolation voltage (in to out)	3,000	V rms	
Isolation voltage (in to base)	1,550	V rms	
Isolation voltage (out to base)	500	V rms	
Output power	150	Watts	
Storage temperature	–40 to +125	°C	
Operating temperature	–20 to +100	°C	Baseplate. See Fig. 6
Pin soldering temperature	500 (260)	°F (°C)	<5 sec
Mounting torque	10 (1.13)	in-lbs (N-m)	6 each, 4-40 screw

Thermal Resistance and Capacity

Parameter	Min	Typ	Max	Unit
Baseplate to sink; flat, greased surface		0.24		°C/Watt
Baseplate to sink; Grafoil®		0.3		°C/Watt
Baseplate to ambient		15		°C/Watt
Baseplate to ambient; 1000 LFM		2.7		°C/Watt
Thermal capacity		48		Watt-sec/°C

"GRAFOIL" is a registered trademark of Union Carbide Corporation.

ELECTRICAL CHARACTERISTICS

Electrical characteristics apply over the full operating range of input voltage, output load (resistive) and baseplate temperature, unless otherwise specified. All temperatures refer to the operating temperature at the center of the baseplate.

MODULE OPERATING SPECIFICATIONS

Parameter	Min	Typ	Max	Unit	Notes
Operating input voltage	36	48	75	Vdc	See Fig. 1
Input surge withstand			100	Vdc	<100ms. See Fig. 1
Output voltage setpoint	11.9	12.0	12.1	Vdc	48V in; full load; 25°C
Output OVP setpoint	14.0	14.3	14.5	Vdc	25°C; recycle input voltage to restart
Output power	0		150	Watts	
Efficiency	84.5	86		%	48V in; 75% of full load; 25°C. See Fig. 5

MODULE INPUT SPECIFICATIONS

Parameter	Min	Typ	Max	Unit	Notes
Undervoltage turn-on		35	36	Vdc	See Fig. 1
Undervoltage turn-off	32	33		Vdc	See Fig. 1
Overvoltage turn-off	76	79	81	Vdc	See Fig. 1
Overvoltage turn-on	75	77		Vdc	See Fig. 1
Input reflected ripple current, p-p		50	65	mA	48V in, full load, AC current probe, 20MHz BW
Inrush charge		120×10^{-6}	150×10^{-6}	Coulombs	48V in
Ripple rejection at 120Hz	52	57		dB	48V in, 75W load. See Fig. 10
Ripple rejection at 2,400Hz	35	40		dB	48V in, 75W load. See Fig. 10
Dissipation, standby		3.0		Watts	48V in, no load
Dissipation, disabled		0.02	0.03	Watts	48V in, PC grounded

MODULE OUTPUT SPECIFICATIONS

Parameter	Min	Typ	Max	Unit	Notes
Line regulation		±0.02	±0.1	%	Input voltage: 36 to 75V; full load
Load regulation		±0.05	±0.1	%	No load to full load, 48V in
Temperature regulation		±0.002	±0.005	% / °C	-20 to +100°C
Ripple and noise, p-p, 20MHz BW		210	260	mV	48V in, full load. See Fig. 4
Crossover frequency	35	45		kHz	See Fig. 8
Phase margin	55	72		deg.	See Fig. 8
Dynamic response (25/75/25)					Load change from 25% to 75% to 25% of rating (0.7A/µs, 25°C)
Peak deviation		1.5	2.0	%	Of output voltage setpoint
Settling time		80	150	µs	To within 1% of output voltage. See Fig. 9
Current limit	102	115	130	% of I max.	Output voltage 95% of nominal
Short circuit current	100	112	125	% of I max.	Output voltage <250mV
Power sharing accuracy		±2	±5	%	10 to 100% of full load
Programming range	10		110	%	Of nominal output voltage. See Figs. 18, 19, 22

MODULE CONTROL SPECIFICATIONS

Parameter	Min	Typ	Max	Unit	Notes
PRIMARY SIDE (PC = Primary Control; PR = Parallel)					
PC bias voltage	5.7	5.9	6.1	Vdc	PC current = 1.0 mA
current limit	2.2	3.0	4.0	mA	PC voltage = 5.6 Volts
PC module disable	2.5	2.6	2.7	Vdc	Must be able to sink ≥4mA. See Fig. 12
PC module enable delay		4	7	ms	See Fig. 2
PC module alarm			0.5	Vavg	UV, OV, OT, module fault. See Figs. 13, 15
PR emitter amplitude	5.7	5.9	6.1	Volts	PR load >30 ohms, < 30 pF
PR emitter current	150			mA	
PR receiver impedance	375	500	625	Ohms	25°C
PR receiver threshold	2.4	2.5	2.6	Volts	Minimum pulse width: 20ns
PR drive capability	12			Modules	Without PR buffer amplifier
SECONDARY SIDE (SC = Secondary Control)					
SC bandgap voltage	1.22	1.23	1.24	Vdc	Referenced to –Out
SC resistance	990	1000	1010	Ohms	
SC capacitance		0.033		µF	
SC module alarm		0		Vdc	With open trim; referenced to –Out. See Fig. 17

MODULE GENERAL SPECIFICATIONS

Parameter	Min	Typ	Max	Unit	Notes
Isolation voltage (in to out)	3,000			V rms	Complies with reinforced insulation requirements
Isolation voltage (in to base)	1,550			V rms	Complies with basic insulation requirements
Isolation voltage (out to base)	500			V rms	Complies with operational insulation requirements
Isolation capacitance (in to out)		10		pF	Three terminal measurement
Isolation capacitance (in to base)		200		pF	Three terminal measurement
Isolation capacitance (out to base)		200		pF	Three terminal measurement
Isolation resistance (in to out)		10		Megohms	
Weight		2.1 (60)		Ounces (grams)	
Temperature limiting	100	115		°C	See Figs. 13, 15
Agency Approvals		UL, CSA, TÜV, VDE, BABT, CE			IEC 950. With a 7A fuse in series with the +Input In process as of Nov. 1, 1996

OPERATING CHARACTERISTICS

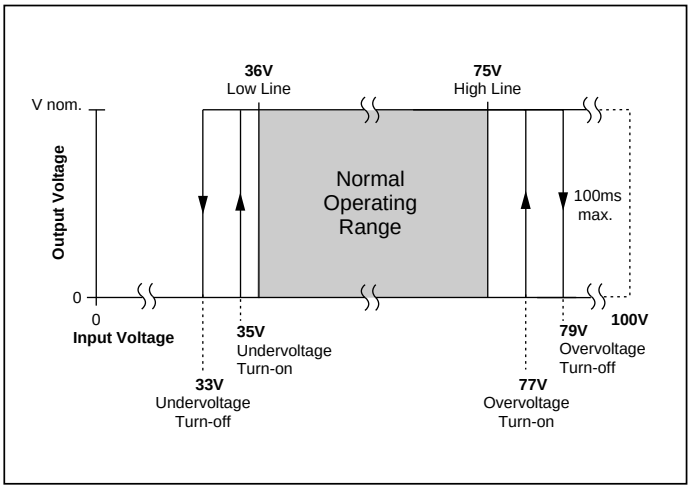


Figure 1*—Input voltage range (typical).*

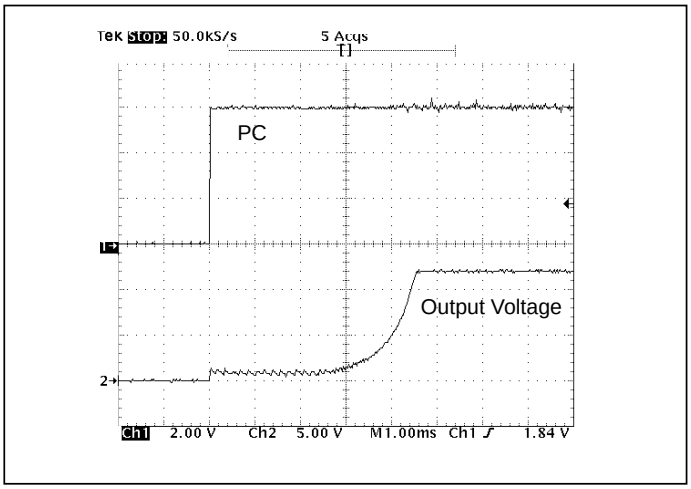


Figure 2*—Turn-on characteristics—Output voltage from PC enable, 48V in, 75W load. See Figure 12.*

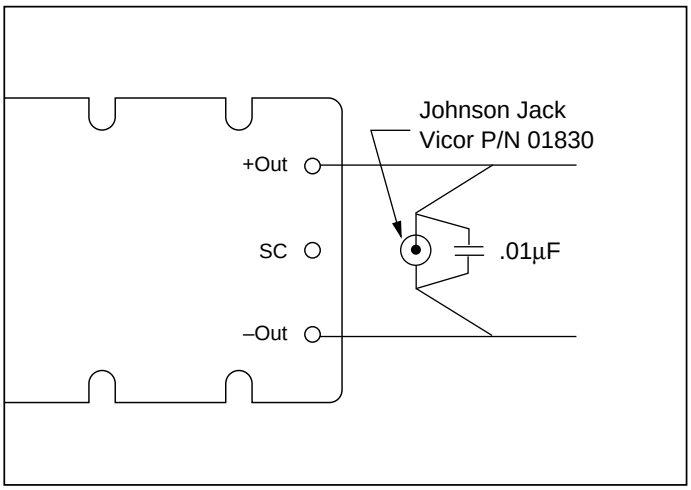


Figure 3*—Output ripple and transient response test circuit.*

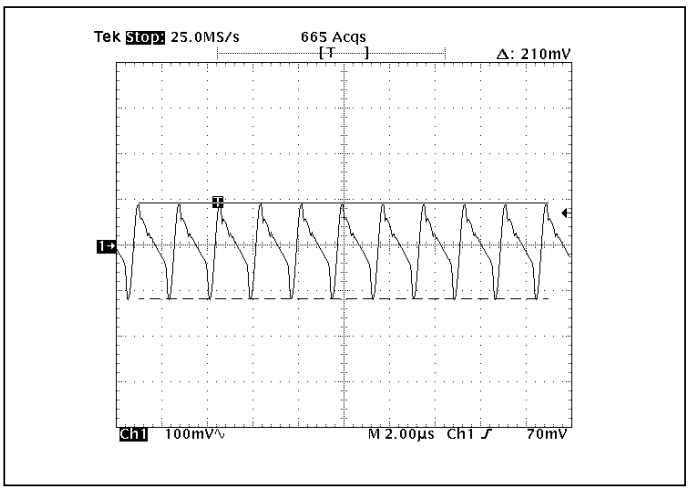


Figure 4*—Output ripple and noise 48V in, 150W load.*

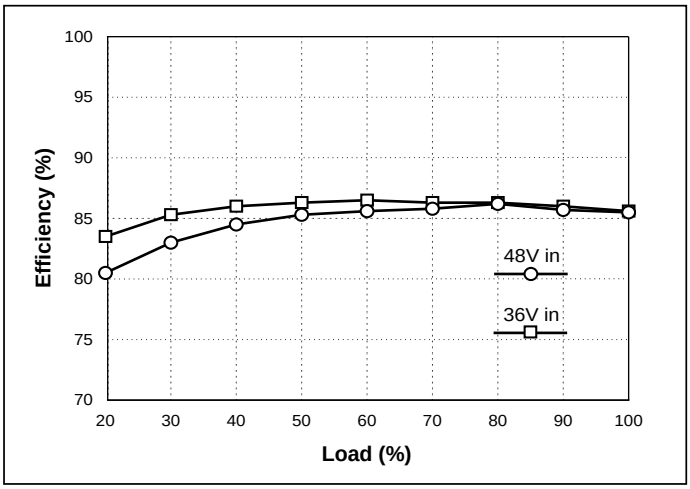


Figure 5*—Efficiency vs. load and input voltage.*

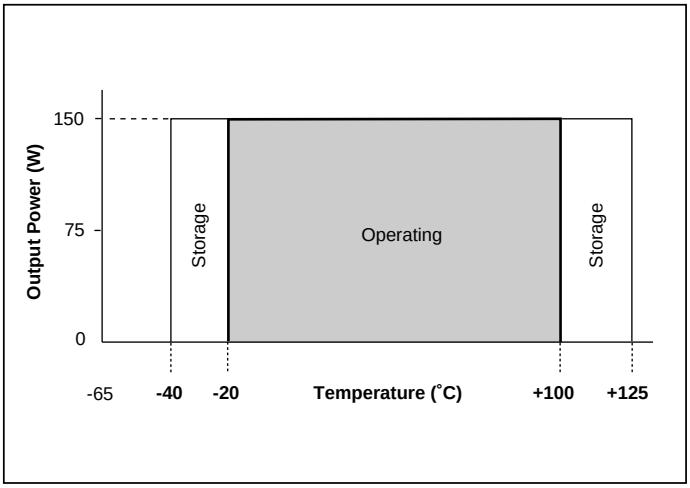


Figure 6*—Power derating.*

OPERATING CHARACTERISTICS, continued

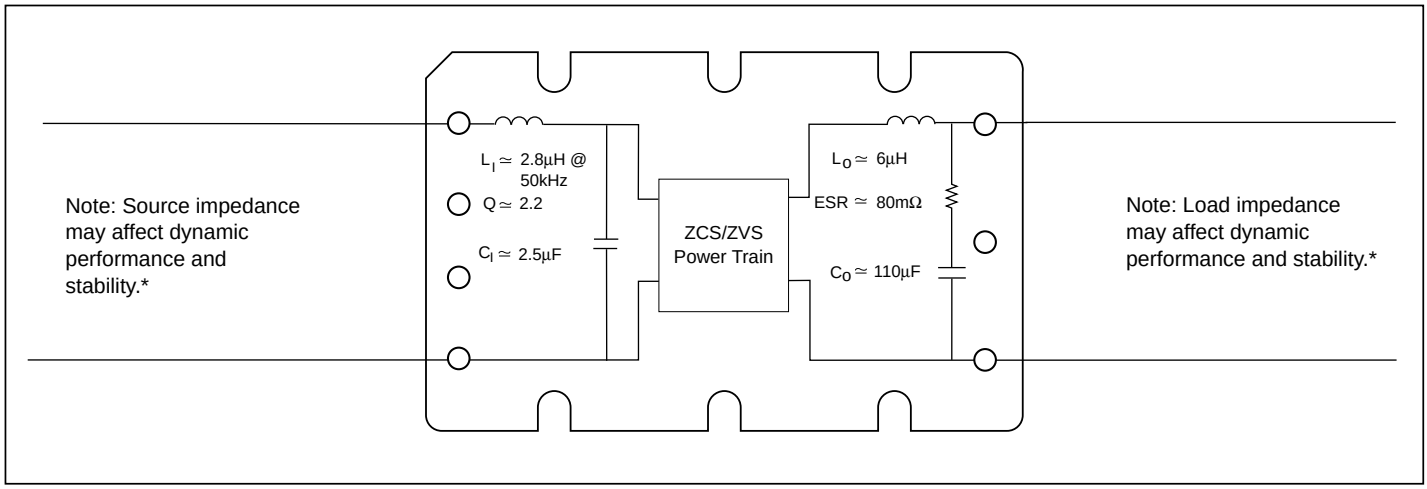


Figure 7*—Internal Input / Output filter elements. *For application support please contact a Vicor Technical Support Center.*

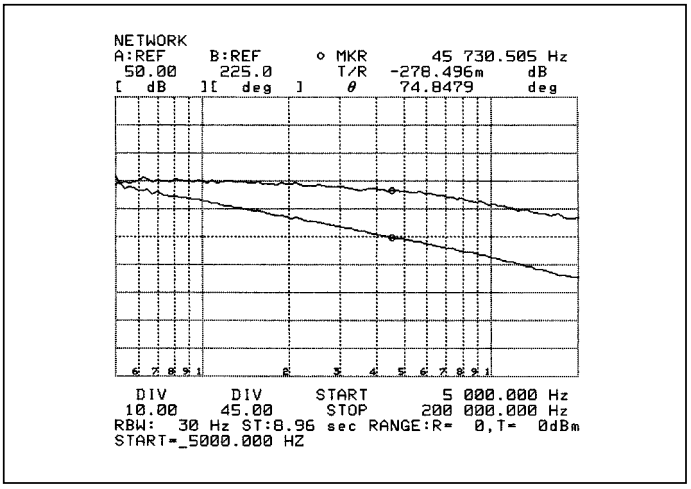


Figure 8*—Loop crossover characteristics, 48V in, 75W load.*

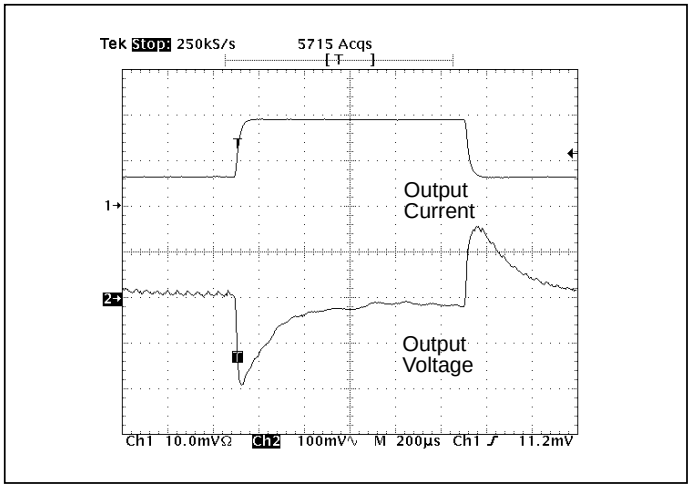


Figure 9*—Transient response, 48V in, 25 - 75 - 25% load.*

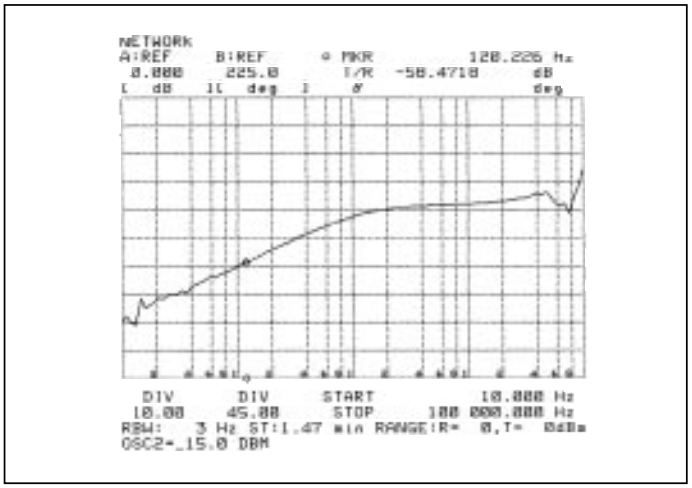


Figure 10*—Input rejection: Vout / Vin, 48V in, 75W load.*

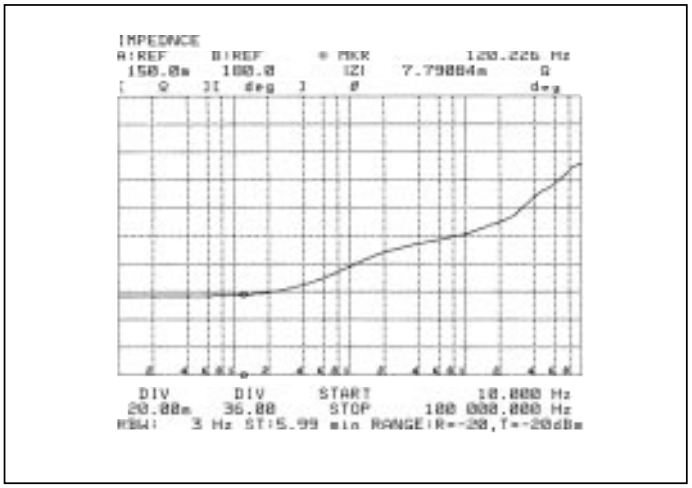


Figure 11*—Output impedance: Vout / Iout, 48V in, 75W load.*

CONTROL FUNCTIONS - PC PIN

Module Enable/Disable

The module may be disabled by pulling PC below 2.5V with respect to the –Input. This may be done with an open collector transistor, relay, or optocoupler. Multiple converters may be disabled with a single transistor or relay either directly or via "OR'ing" diodes. See Figure 12.

Primary Auxiliary Supply

At 5.9V, PC can source up to 2.2mA. In the example shown in Figure 14, PC powers a "green" Module Enabled LED.

Module Alarm

The module contains "watchdog" circuitry which monitors input voltage, operating temperature and internal operating parameters. In the event that any of these parameters are outside of their allowable operating range, the module will shut down and PC will go low. PC will periodically go high and the module will check to see if the fault (as an example, overtemperature) has cleared. If the fault has not been cleared, PC will go low again and the cycle will restart. The SC pin will go low in the event of a fault and return to its normal state after the fault has been cleared. See Figures 13, 15.

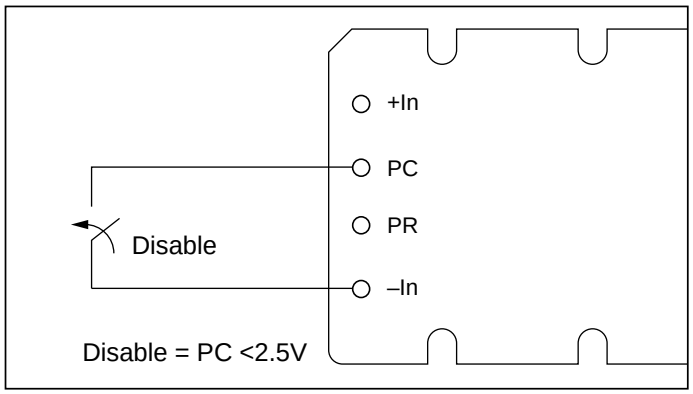


***Figure 12**—Module enable/disable.*

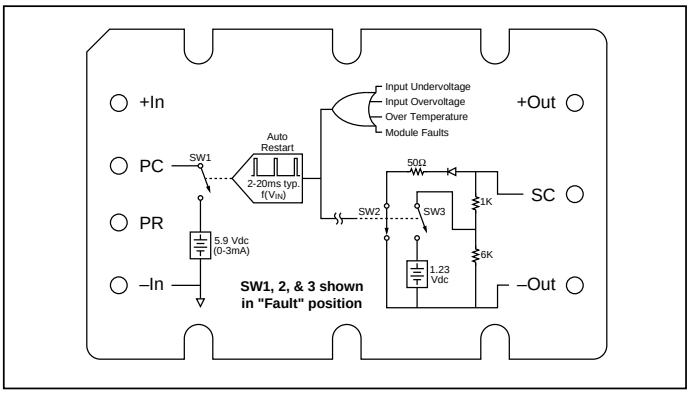


***Figure 13**—PC/SC module alarm logic.*

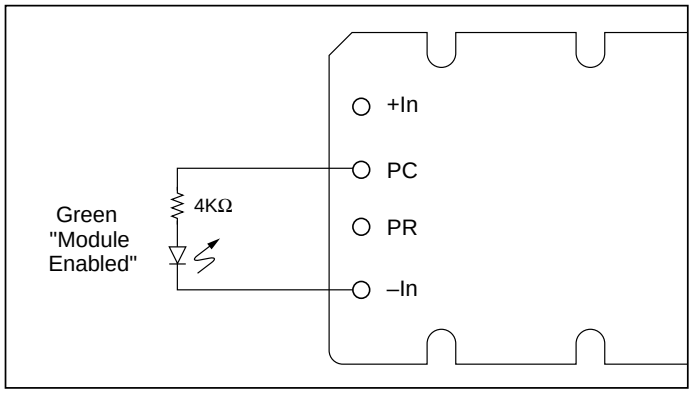


***Figure 14**—LED on-state indicator.*

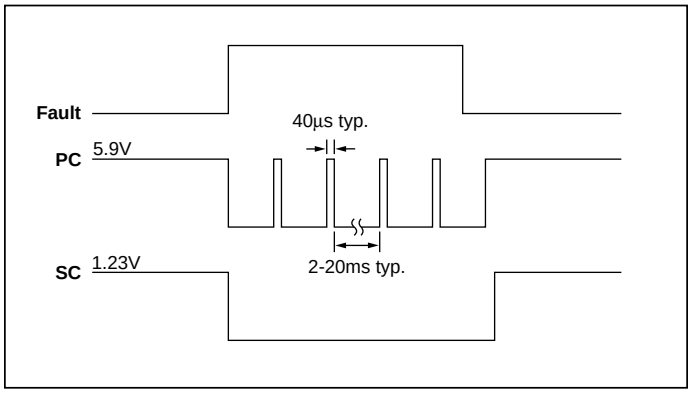


***Figure 15**—PC/SC module alarm timing.*

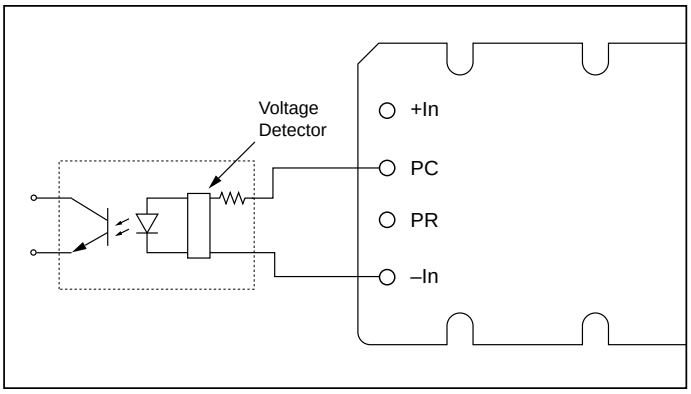


***Figure 16**—Isolated on-state indicator.*

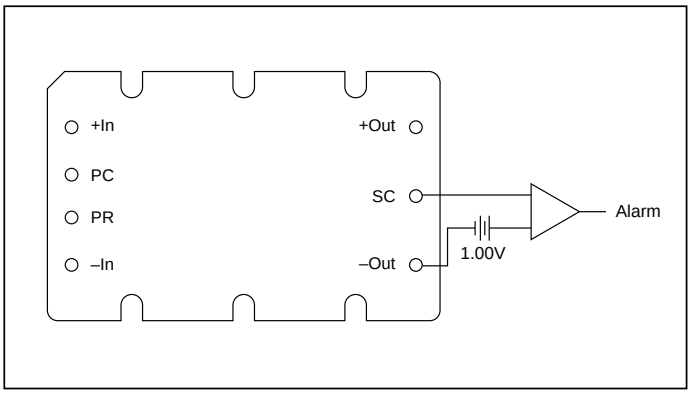


***Figure 17**—Secondary side on-state indicator.*

CONTROL FUNCTIONS - SC PIN

Output Voltage Programming

The output voltage of the converter can be adjusted or programmed via fixed resistors, potentiometers or voltage DACs. See Figures 18, 19, 20, 21, 22, 23.

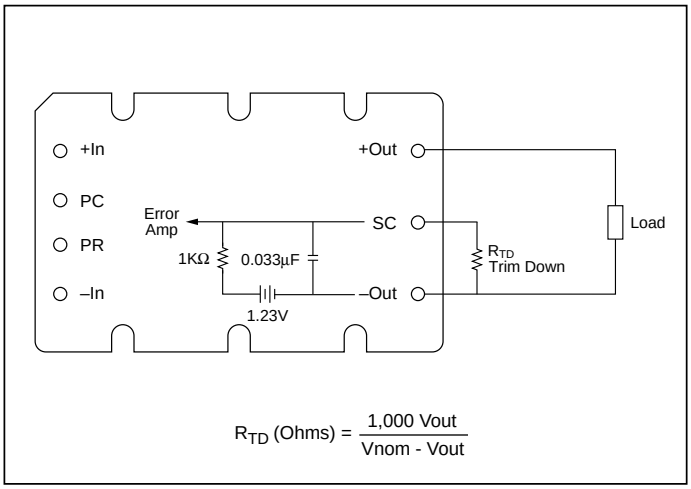


***Figure 18**—Output voltage trim down circuit.*

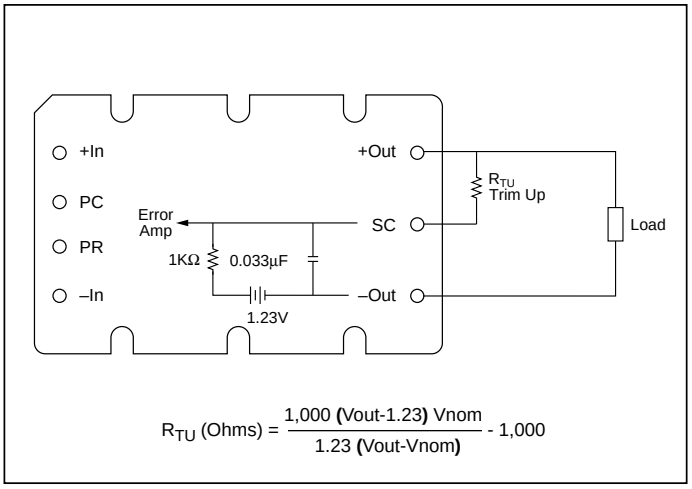


***Figure 19**—Output voltage trim up circuit.*

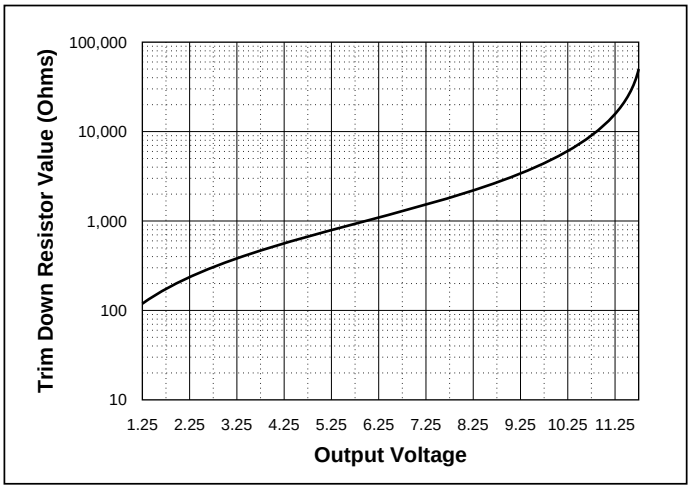


***Figure 20**—Trim down resistor value (RTD).*

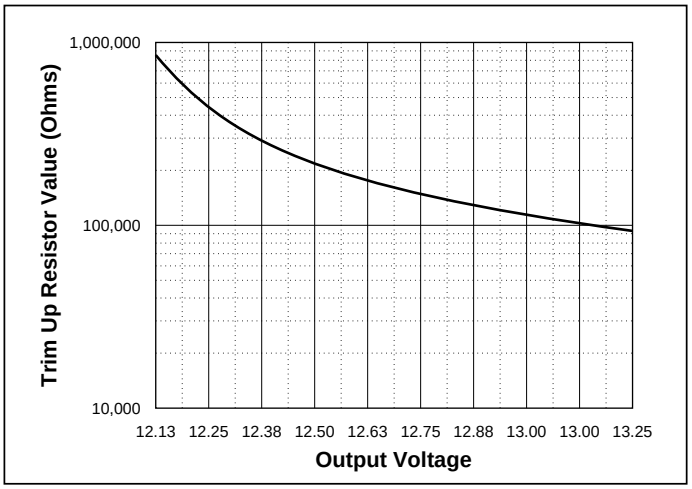


***Figure 21**—Trim up resistor value (RTU).*

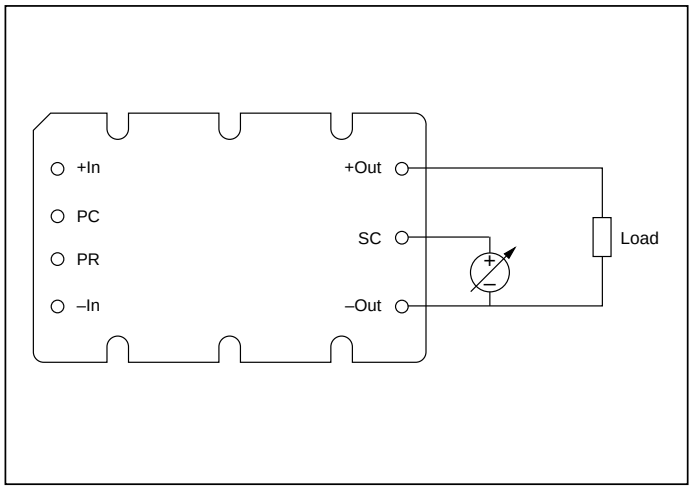


***Figure 22**—Output Voltage Programming.*

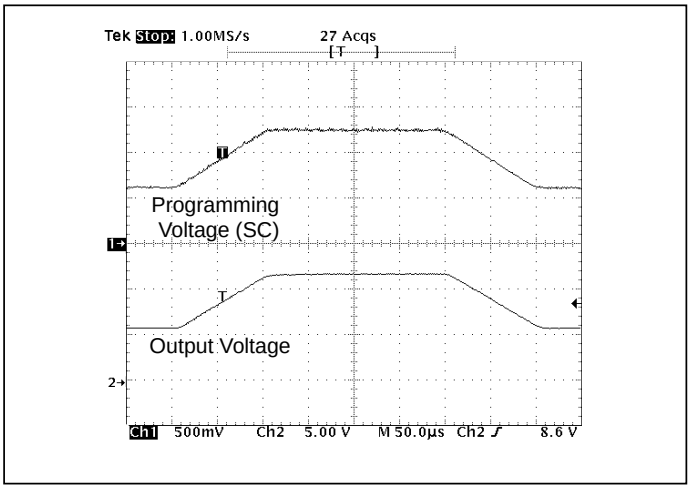


***Figure 23**—Programming Response (12V - 6V - 12V) at 75W load.*

CONTROL FUNCTIONS - PR PIN

Parallel Operation

The PR pin supports paralleling for increased power with N+1 (N+M) redundancy and phased array capability. Modules of the same input voltage, output voltage, and power level will current share if all PR pins are suitably interfaced.

Compatible interface architectures include the following:

DC coupled single-wire interface. All PR pins are directly connected to one another. This interface supports current sharing but is not fault tolerant. Minus In pins must be tied to the same electric potential. See Figure 24.

AC coupled single-wire interface. All PR pins are connected to a single communication bus through 330pF (500V) capacitors. This interface supports current sharing and is fault tolerant except for the communication bus. See Figure 24.

Fault tolerant interface (FTI). Each PR pin is interfaced to every other PR pin. This interface supports current sharing and is fault tolerant. See Figures 25, 30, and 32.

Phased Array Controller (PAC) Interface. All PR and PC pins are connected to a PAC chip. See Figures 26, 30, and 32. The PAC chip supports current sharing and automatically adjusts the relative phase of each module in the array in increments of 360°/N, where N is the number of enabled modules within the array. See Figures 29 and 31. Phasing greatly reduces the filtering requirements of high power, multiple module arrays.

Transformer coupled interface. Modules or arrays of modules may also be interfaced to share a load while providing galvanic isolation between PR pins via a transformer coupled interface. For large arrays, buffering may be required. The power source for the buffer circuit may be derived from PC pins. Please contact a Vicor Technical Support Center.

Module Paralleling Architectures and Features

Architecture	Current sharing	Module fault tolerant	Module and bus fault tolerant	Safety isolated interface	Phased power processing
DC coupled single wire	✔				
AC coupled single wire	✔	✔			
FTI	✔	✔	✔		
PAC	✔	(1)	(1)		✔
Transformer coupled	✔	✔		✔	

(1) Consult factory about fault tolerant phased arrays.

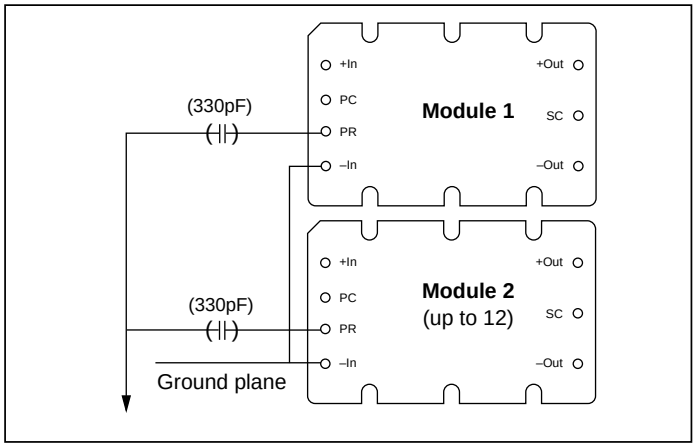


Figure 24*—DC (AC) coupled single-wire interface. (Care should be taken to minimize interconnect parasitic capacitance and inductance).*

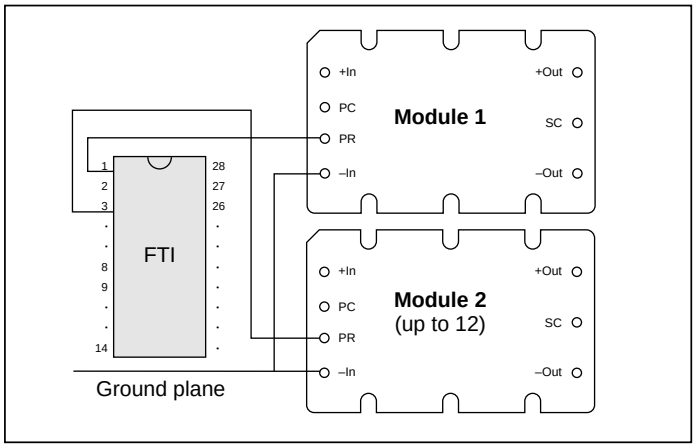


Figure 25*—Redundant array with fault tolerant interface.*

FTI Part Numbers: DIP: FTI-12D
SOIC: FTI-12S

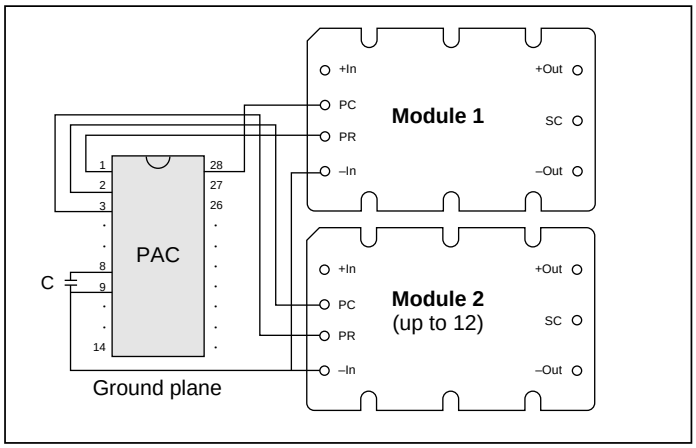
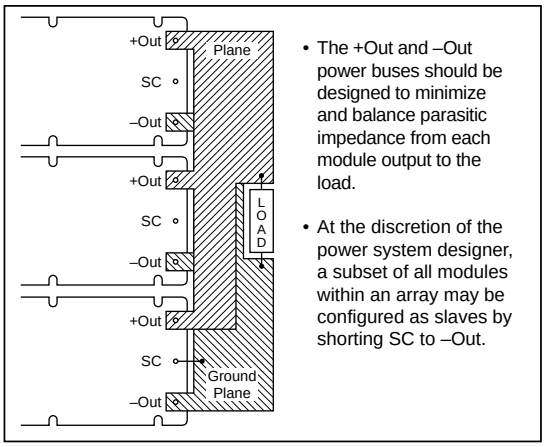


Figure 26*—Redundant array with phased array controller interface.*

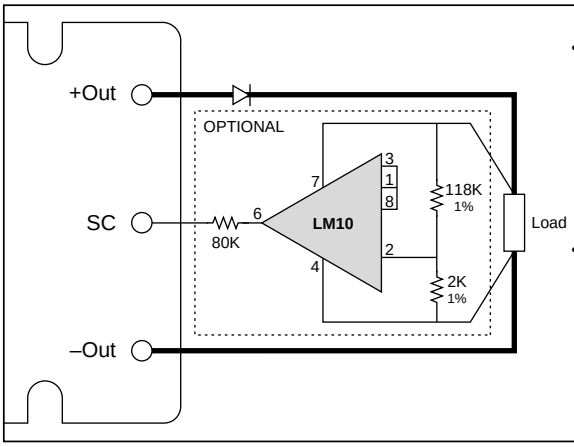
PAC Part Numbers: DIP: PAC-12D
SOIC: PAC-12S

CONTROL FUNCTIONS, continued



- The +Out and –Out power buses should be designed to minimize and balance parasitic impedance from each module output to the load.
- At the discretion of the power system designer, a subset of all modules within an array may be configured as slaves by shorting SC to –Out.

Figure 27*—N+1 module array output connections.*



- OR'ing diodes may be inserted in series with the +Out pin of each module to provide output fault tolerance. As an example, the circuit shown may be used to compensate for the voltage drop of the diode.
- This module is designed for point of load regulation, where remote sensing is not required. Active compensation, as shown here, may be used in applications with significant distribution losses.

Figure 28*—OR'ing diode connections and voltage drop compensation.*

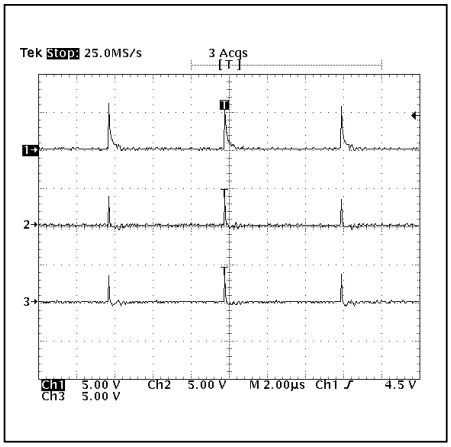
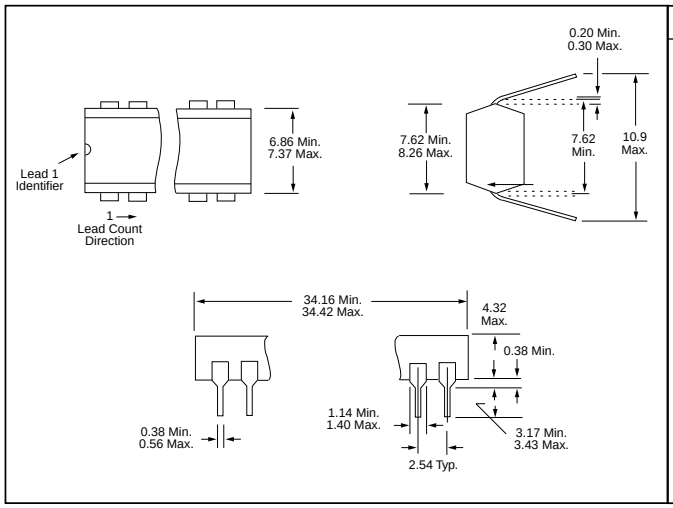


Figure 29*—PR waveforms of a 3 module array with FTI.*



PIN #	FTI	PAC
1	PR1	PR1
2	NC	PC2
3	PR2	PR2
4	NC	PC3
5	PR3	PR3
6	NC	PC4
7	PR4	PR4
8	NC	Vcc
9	NC	Gnd
10	NC	PC5
11	PR5	PR5
12	NC	PC6
13	PR6	PR6
14	NC	PC7
15	PR7	PR7
16	NC	PC8
17	PR8	PR8
18	NC	PC9
19	PR9	PR9
20	NC	PC10
21	PR10	PR10
22	NC	OK
23	NC	SYS RES
24	NC	PC11
25	PR11	PR11
26	NC	PC12
27	PR12	PR12
28	NC	PC1

Figure 30*—Mechanical drawing of DIP package (FTI-12D, PAC-12D).*

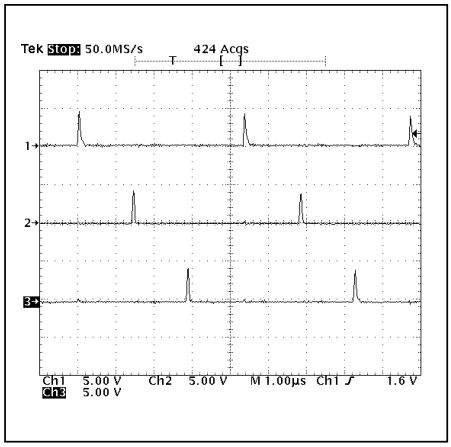
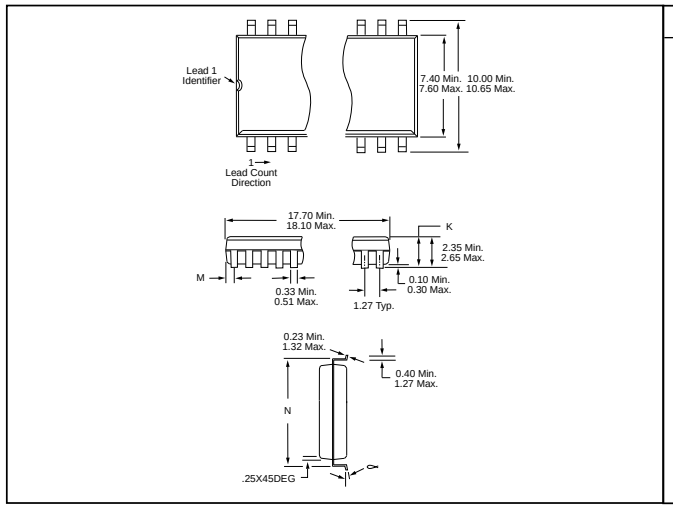


Figure 31*—PR waveforms of a 3 module array with PAC.*

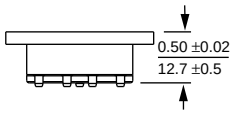


PIN #	FTI	PAC
1	PR1	PR1
2	NC	PC2
3	PR2	PR2
4	NC	PC3
5	PR3	PR3
6	NC	PC4
7	PR4	PR4
8	NC	Vcc
9	NC	Gnd
10	NC	PC5
11	PR5	PR5
12	NC	PC6
13	PR6	PR6
14	NC	PC7
15	PR7	PR7
16	NC	PC8
17	PR8	PR8
18	NC	PC9
19	PR9	PR9
20	NC	PC10
21	PR10	PR10
22	NC	OK
23	NC	SYS RES
24	NC	PC11
25	PR11	PR11
26	NC	PC12
27	PR12	PR12
28	NC	PC1

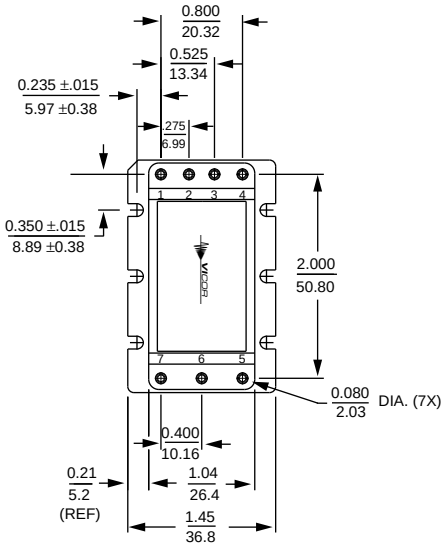
Figure 32*—Mechanical drawing of SOIC package (FTI-12S, PAC-12S).*

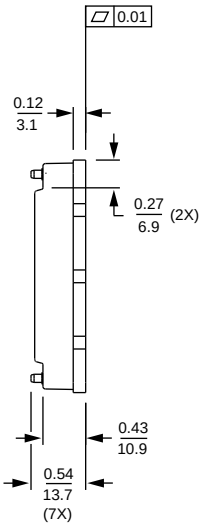
MECHANICAL DRAWINGS

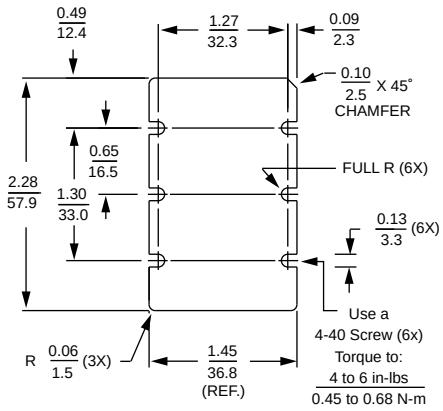
MODULE OUTLINE



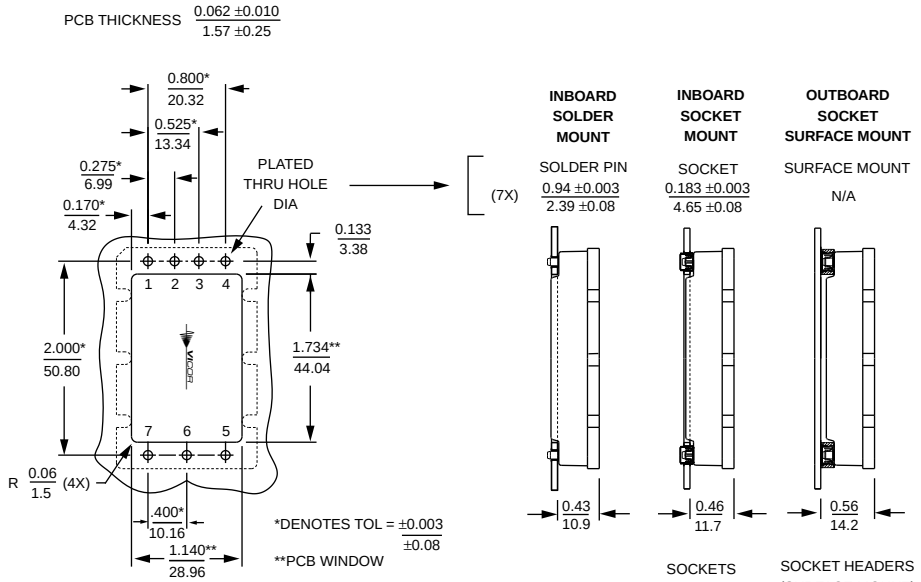
Converter Pins		
No.	*Function*	*Label*
1	+In	+
2	Primary Control	PC
3	Parallel	PR
4	–In	–
5	–Out	–
6	Secondary Control	SC
7	+Out	+

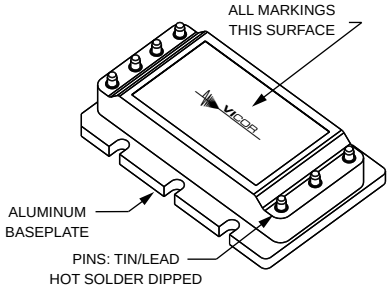






PCB MOUNTING SPECIFICATIONS





Unless otherwise specified, dimensions are in inches / mm

Decimals	Tol.	Angles
0.XX	±0.01 / ±0.25	±1°
0.XXX	±0.005 / ±0.127	

NOTES

Vicor U.S.A.
23 Frontage Road
Andover, MA, 01810
Tel: 800-735-6200
Fax: 508-475-6715

Lombard, IL
Tel: 630-769-8780
Fax: 630-769-8782

Sunnyvale, CA
Tel: 408-522-5280
Fax: 408-774-5555

Vicor France
Tel: +33-1-3452-1830
Fax: +33-1-3452-2830
Free Phone:
0800 419 419

Vicor Germany
Tel: +49-89-962-439-0
Fax: +49-89-962-439-39
Free Phone:
0130 18 29 18

Vicor Hong Kong
Tel: +852-2956-1782
Fax: +852-2956-0782

Vicor Italy
Tel: +39-2-2247-2326
Fax: +39-2-2247-3166

Vicor Taiwan
Tel: +886-2-708-8020
Fax: +886-2-755-5578

Vicor U.K.
Tel: +44-1276-678-222
Fax: +44-1276-681-269

Vicor Integration Architects

Aegis Power Systems
Murphy, NC
Tel: 704-837-4029
Fax: 704-837-4106

ConverTec Corporation
Roseville, MN
Tel: 612-604-0289
Fax: 612-604-0293

Freedom Power Systems
Austin, TX
Tel: 512-833-6177
Fax: 512-833-6181

Mission Power Solutions
Oceanside, CA
Tel: 619-631-6846
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E-Mail to: Vicor Express: vicorexp@vicr.com
Technical Support: apps@vicr.com

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